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BCA (Sem-I) – Computer

System Architecture (102) Core-II

2024

Time : 3 hours

Full Marks : 70

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Answer ***all*** *sections as directed.*

Section–A

Objective type Questions

Compulsory

1. Choose the correct answer of the following multiple choice questions : 2×10=20

(i) A Flip-Flop is used for:

a) Arithmetic operations

b) Storing a single bit of data

c) Managing memory access

d) Controlling input devices

(ii) Which type of memory is used for temporary storage of data during processing?

a) ROM

b) RAM

c) Hard Disk

d) Optical Disk

(iii) Which of the following is the main function of the ALU?

a) Perform arithmetic operations

b) Control data flow

c) Store instructions

d) Manage memory

(iv) Choose which of the following is/are in SOP form:

a) $(X'+Y)(W'+Z')$

b) $X'Y+W'Z'$

c) Both the above

d) None of these

(v) $F=\Pi(2,3,4,5)$ is equivalent to:

a) $F=m_2.m_3.m_4.m_5$

b) $F=M_2.M_3.M_4.M_5$

c) $F=M_2+M_3+M_4+M_5$

d) None of these

(vi) How many selection lines in 2^n-to-1 multiplexer:

a) N+1

b) N

c) N – 1

d) None of these

(vii) Which of the following is not a combinational circuit:

a) multiplexer

b) decoder

c) encoder

d) flip flop

(viii) Product of all literals are known as:

a) minterms

b) maxterms

c) both (a) & (b)

d) none of these

(ix) Encircling four adjacent 1's in K-Map is known as:

a) Pair

b) Octet

e) Quad

d) None of these

(x) Which component generates the DMA request signal?

a) CPU

b) Memory

c) I/O devices

d) Cache

Section–B

Short Answer type Questions

Answer any **four** questions of the following : 5×4=20

2. Define the following terminology:

(a) Combinational circuit

(b) Sequential circuit

(c) Encoder

(d) Multiplexer

(e) Minterm

3. Define universal gate. How to obtain NAND gate using NOR gate.

4. Prove the following expressions by algebraic method:

(a) $X'Y' + YZ + X'YZ' = X' + YZ$

(b) $(X'Y + XY')' = XY + X'Y'$

5. Draw circuit diagram of the following Boolean expressions:

(a) $(XY)' + (YZ)' + (XZ)'$

(b) $(A+B)'.(BC)'.(AC)'$

6. Write similarities and differences between SR and JK flip flop.

7. Suppose A, B and C are three inputs and S(Sum) and C(Carry) are two outputs in a

full adder. Write the truth table for this full adder and derive the boolean expressions for S and C.

Section–C

Long Answer type Questions

Answer any **two** questions of the following : 15×2=30

8. Construct a 4-to-16 line decoder using logic gates. Provide its truth table and draw the corresponding circuit diagram.

9. Represent 300.5 using the IEEE 754 standard for 32-bit floating point representation.

10. Simplify the following Boolean function F together with the don't care condition d in

(i) sum-of-product form and

(ii) product-of-sum form

$$F(A,B,C,D)=\sum(0,2,3,6,7)$$

$$d(A,B,C,D)=\sum(8,10,11,15)$$

11. Write the short notes on the following:

(a) DeMorgan's theorem

(b) DMA

(c) K-Map

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